Retrofit Radar

Predicting building energy use to decide which buildings to retrofit first

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Which buildings should be fixed first?

- Buildings use about a third of the world’s energy and energy-related CO_2.
- Retrofits (insulation, heat pumps, HVAC upgrades) cut that, but they are expensive, so organisations must choose.
- Raw energy use misleads: a hospital always uses more than a warehouse, and a building in a cold state always heats more.

What we need: how much a building should use, given its type, size, age and weather.

Our approach: actual vs expected energy

1. Predict

A model learns each building's site EUI from its type, size, age, state and weather.

2. Benchmark

A second model, without the Energy Star rating, gives each building's expected EUI.

3. Flag and rank

Buildings using 1.5× their expected EUI or more become retrofit candidates, ranked by wasted MWh and CO_2.

Site EUI (energy use intensity) = energy used per square foot per year, in kBtu/ft²/yr. Dividing by floor area means big buildings are not flagged just for being big.

The data: WiDS Datathon 2022

75,757

building-years to train on (Years 1–6, 64 columns)

9,705

buildings to predict (Year 7, no answers given)

56,382

distinct buildings: the same building repeats across years

Cleaning step	Why
Dropped wind and fog columns	54–60% empty in train, 88–94% in test
Removed 12 duplicate rows; build years before 1800 set to missing (25)	Data-entry errors
Filled missing Energy Star and year built with the facility-type median, plus a “missing” flag	Learned from training rows only, so nothing leaks from the test year
36 monthly temperatures → winter minimum and summer maximum	Near-duplicates of each other and of degree days
Modelled log(EUI) instead of EUI	Long right tail: a few buildings exceed 900 kBtu/ft²/yr

How we tested: two traps, one rule

Trap 1: State 6

State 6 is 67% of the training rows but 0% of the test year (Year 7). So every score is also reported with State 6 removed.

Trap 2: memorisation

The same building repeats across years. Buildings the model has seen: RMSE 29.7. New buildings: RMSE 45.6. A random split would flatter the model.

Our rule: tune on Years 1–5, score once on Year 6 (never used for any decision), then retrain on Years 1–6 to predict Year 7. Boosting was tuned with time-series cross-validation: train on Years 1–3 → check Year 4; train on Years 1–4 → check Year 5.

RMSE (root mean squared error): the typical size of a prediction miss, in kBtu/ft²/yr. Lower is better.

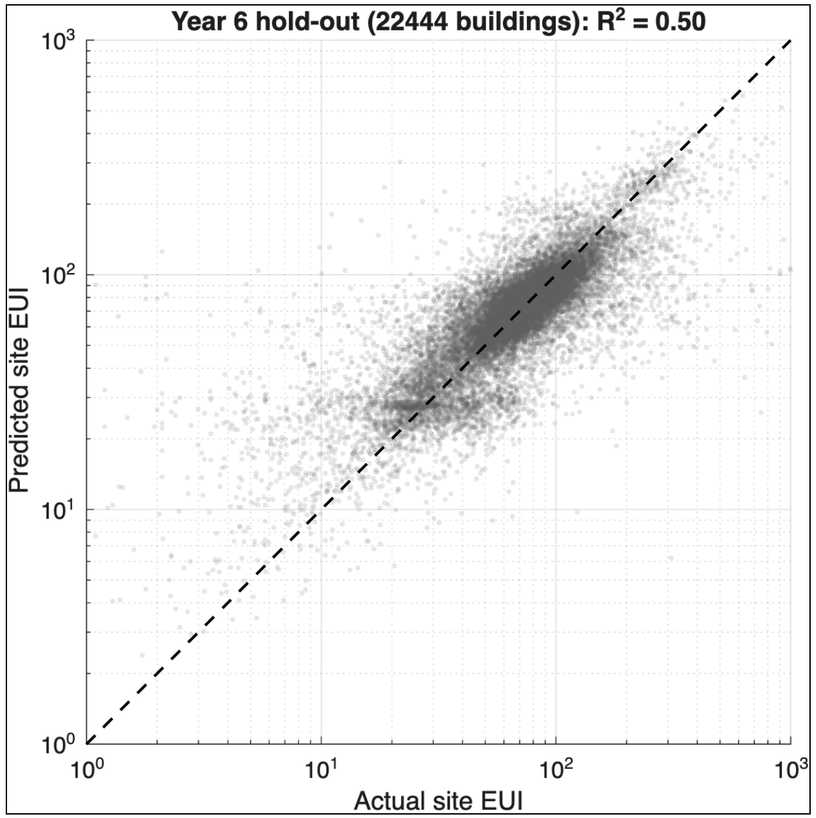
Eight models, all scored on Year 6

#	Model	RMSE	MAE	R^2	RMSE without State 6
0	Facility-type median (baseline)	50.7	27.2	0.21	53.6
1	Linear regression	44.6	23.1	0.39	49.1
2	Regression tree	42.7	21.5	0.44	46.8
3	Bagged trees (100)	42.6	22.7	0.44	44.8
4	Boosted trees, raw target	44.4	22.3	0.39	40.2
5	Boosted trees, log target	40.2	18.8	0.50	42.4
6	Neural network	53.9	29.6	0.11	61.3
7	Tuned boosted trees (final model)	40.3	18.7	0.50	43.2

Biggest gain: modelling log(EUI), model 4 → 5. Tuning confirmed those settings (40.3 vs 40.2).

MAE: average absolute error. R^2: share of the differences between buildings that the model explains (1 = perfect).

Final model: 20% better than baseline



40.3
RMSE on Year 6
(baseline 50.7)

0.50
R² on Year 6
(baseline 0.21)

Model: gradient-boosted trees (LSBoost) on log(EUI): 298 trees, up to 60 splits each, learning rate 0.05, chosen by time-series cross-validation.

Strongest drivers: facility type, Energy Star rating, floor area and year built.

Reading the chart: typical buildings sit on the diagonal; the big misses are extreme users, which the retrofit engine examines next.

Retrofit engine: how a building is scored

1. Expected energy

A benchmark model predicts what the building should use from its type, size, age, state and weather. It ignores Energy Star (which would excuse inefficiency) and never trained on this building.

2. Ratio

Ratio = actual EUI ÷ expected EUI. 1.0 means it uses exactly what similar buildings use; 2.0 means double.

3. Excess energy

(actual − expected EUI) × floor area, converted from kBtu to MWh (× 0.000293).

4. Excess CO_2

Excess MWh × 0.38 t per MWh (assumed grid average).

Example: our top-ranked building

Step	Value
Building	Health care, 3.03M ft²
Expected EUI	192 kBtu/ft²/yr
Actual EUI	342 kBtu/ft²/yr
Ratio	342 ÷ 192 = 1.78
Excess energy	≈ 133,000 MWh/yr
Excess CO_2	≈ 50,700 t/yr

1.78 is between 1.5 and 5, so it is a retrofit candidate: number 1 on the list.

Retrofit engine: three outcomes

Actual ÷ expected	Outcome	Year 6 buildings	What it means
Below 1.5×	Not flagged	19,364	Normal range for similar buildings
1.5× to 5×	Retrofit candidate	2,973 (13.2%)	Uses far more than similar buildings: audit it, biggest excess first
5× or more	Verify meter data	107	Too extreme to be real waste: probably a meter or data error

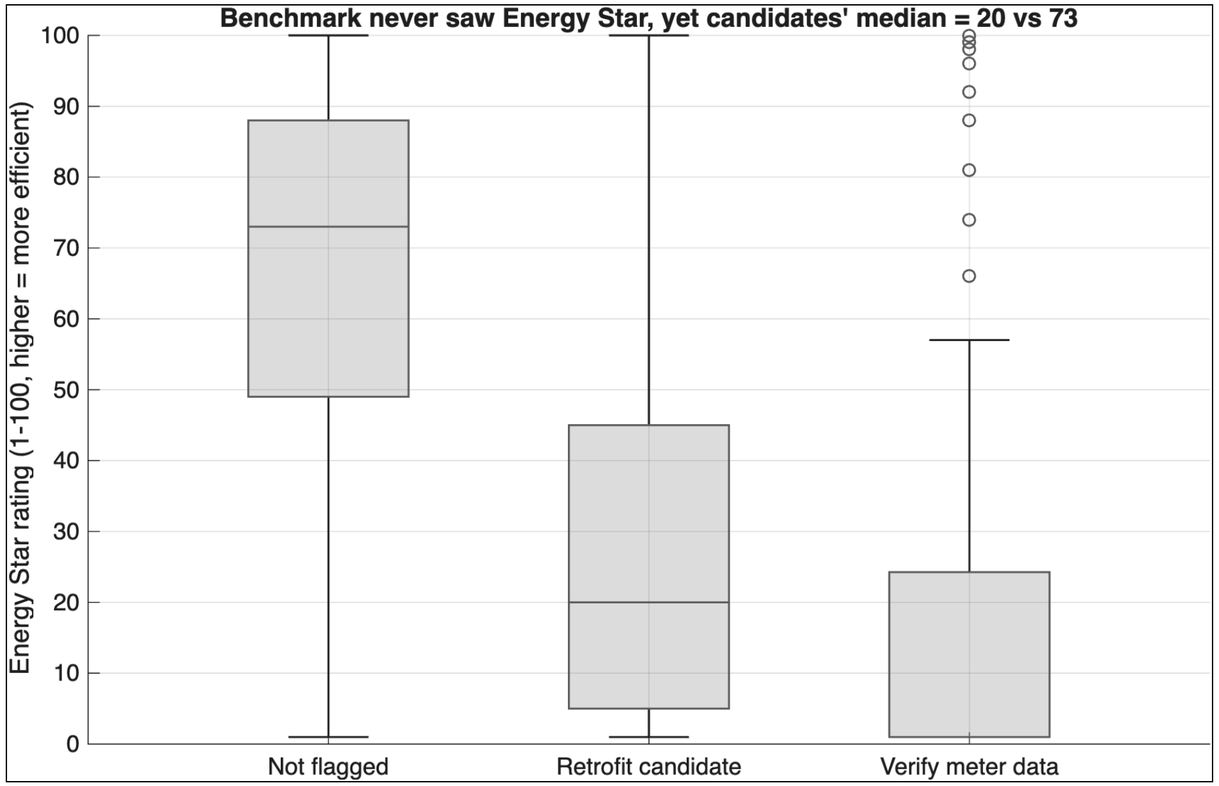
Why 1.5×?

A practical cut-off that flags about one building in eight. It is a policy choice: lower flags more, higher flags fewer, and the app lets you change it.

Why 5×?

Usage that extreme is physically implausible for most building types, so it is more likely a faulty reading than real waste. These are not counted as savings.

Check 1: do the flags match Energy Star?



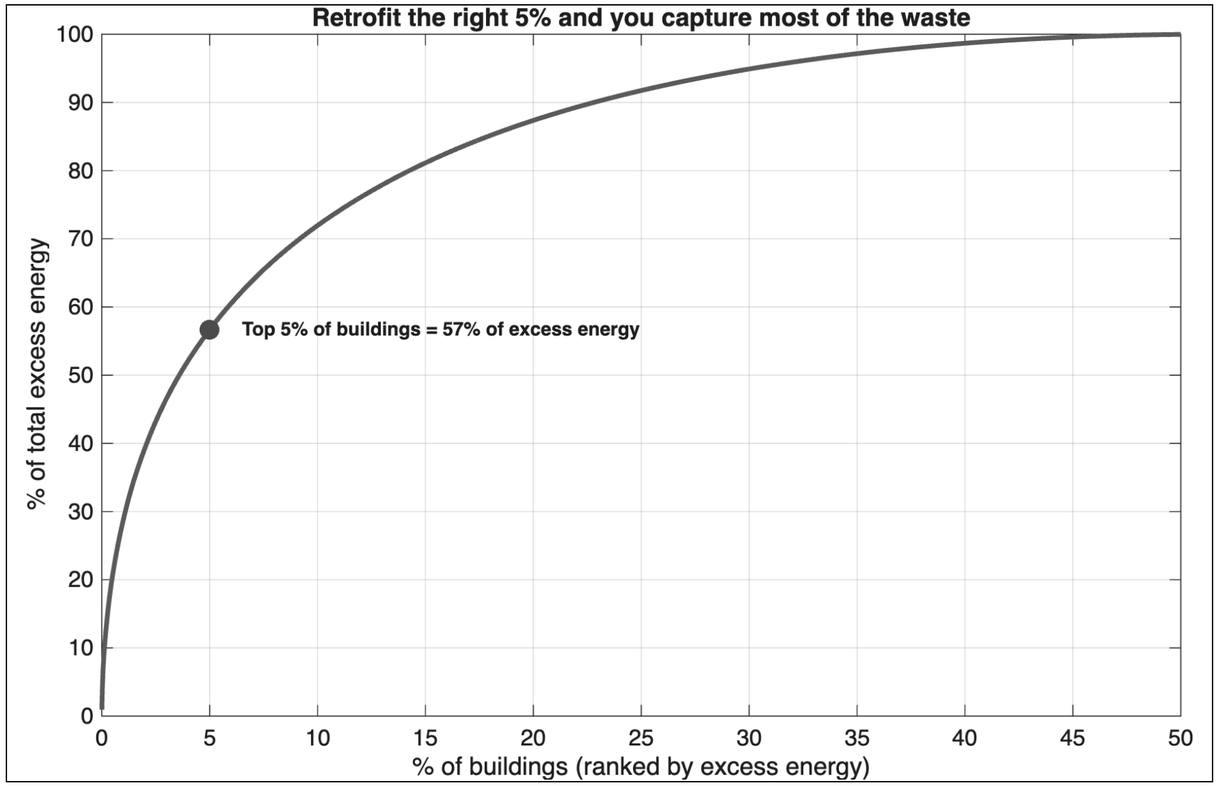
Energy Star rating: a 1–100 efficiency score from the US Environmental Protection Agency; higher means more efficient.

The test: our benchmark never saw this rating. If our flags pick out genuinely inefficient buildings, flagged buildings should have low ratings.

Result: they do. Median rating 20 for retrofit candidates vs 73 for unflagged buildings.

Reading the chart: each box holds the middle half of buildings; the line inside it is the median.

Check 2: where is the waste?



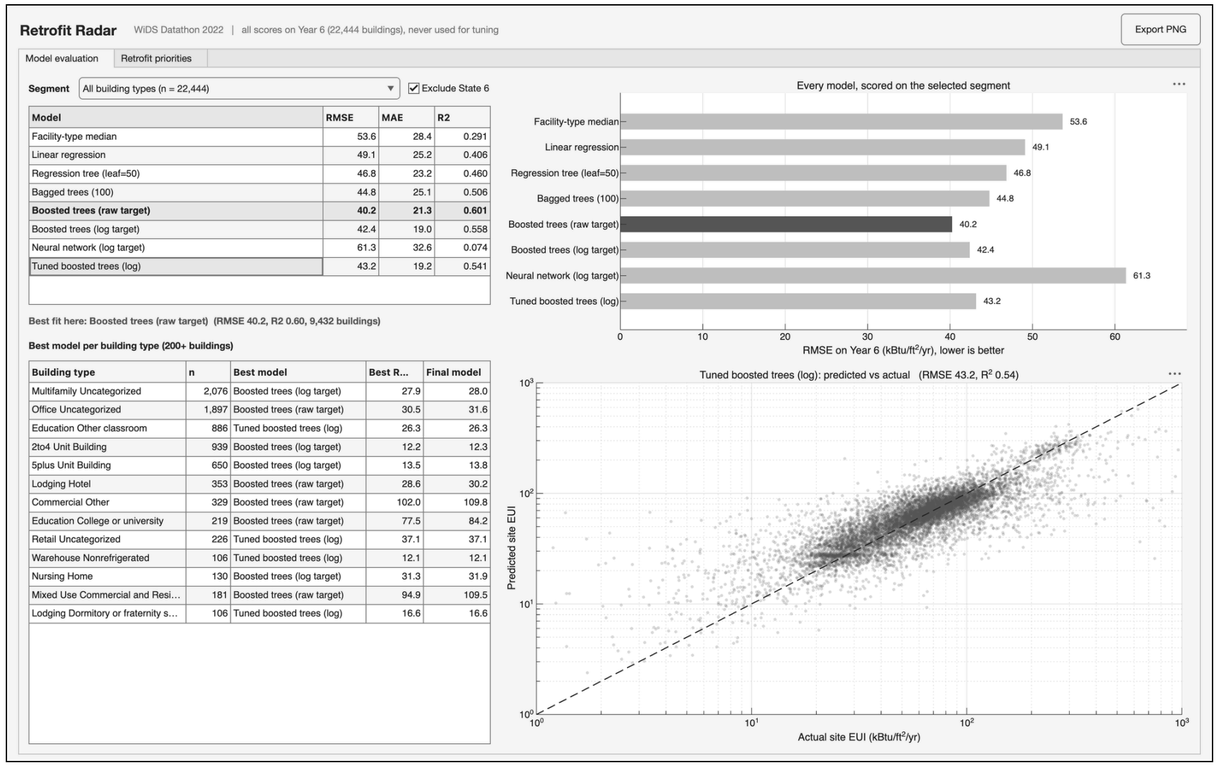
Method: rank buildings by excess energy, largest first, and add up their share of the total excess.

Reading the chart: go along to 5% of buildings, then up to the curve: it reaches 57% of all excess energy.

Result: the top 5% of buildings hold 57% of the waste.

Why it matters: a limited retrofit budget aimed at the top of our list captures most of the opportunity.

MATLAB app, tab 1: model evaluation

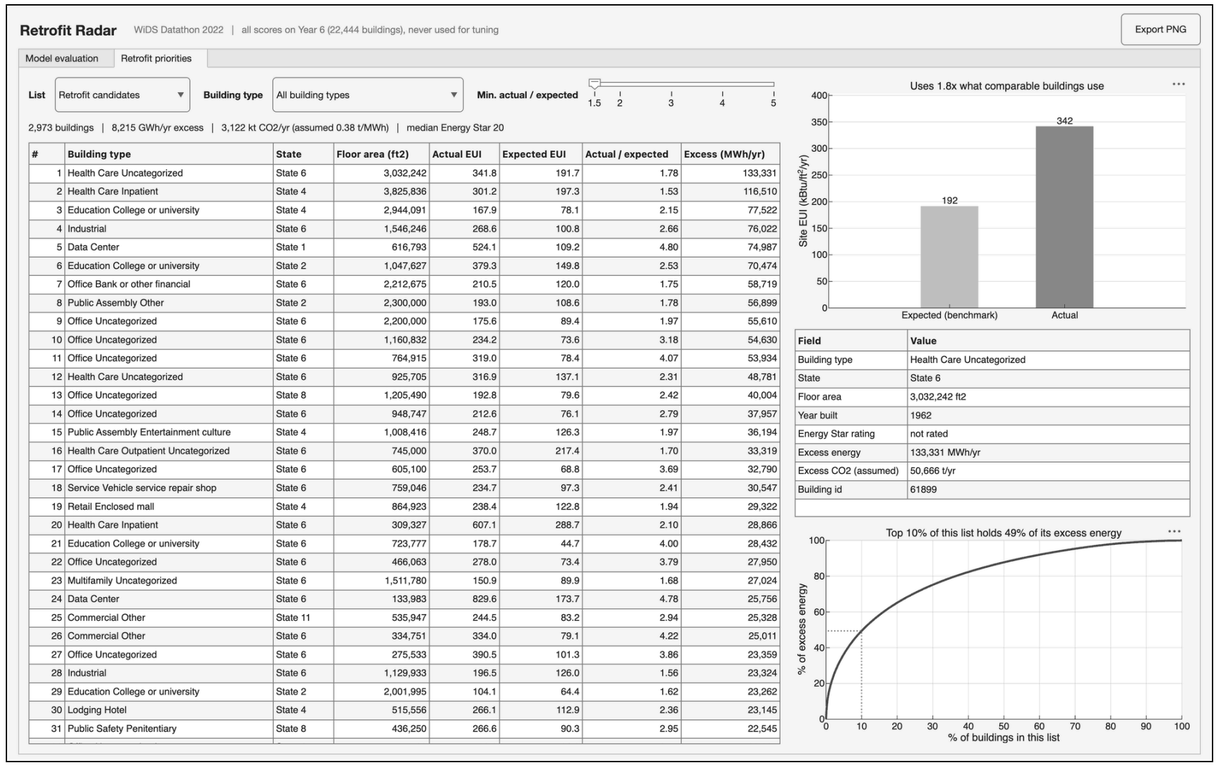


How to use it

1. Pick a building type, or all buildings.
2. Tick “Exclude State 6” to match the test year.
3. All 8 models are re-scored on Year 6; the best is highlighted.
4. Click a model to see its predicted-vs-actual plot.

What it showed: outside State 6, raw-target boosting is best (RMSE 40.2 vs 43.2 for our final model). State 6 is mostly multifamily housing, where the log model is strongest. We kept our pre-chosen model rather than switch after seeing Year 6.

MATLAB app, tab 2: retrofit priorities



#	Building type	State	Floor area (ft2)	Actual EUI	Expected EUI	Actual / expected	Excess (MWh/yr)
1	Health Care Uncategorized	State 6	3,032,242	341.8	191.7	1.78	133,331
2	Health Care Inpatient	State 4	3,825,836	301.2	197.3	1.53	116,510
3	Education College or university	State 4	2,944,091	167.9	78.1	2.15	77,522
4	Industrial	State 6	1,546,246	268.6	100.8	2.66	76,022
5	Data Center	State 1	616,793	524.1	109.2	4.80	74,987
6	Education College or university	State 2	1,047,627	379.3	149.8	2.53	70,474
7	Office Bank or other financial	State 6	2,212,675	210.5	120.0	1.75	58,719
8	Public Assembly Other	State 2	2,300,000	193.0	108.6	1.78	56,899
9	Office Uncategorized	State 6	2,200,000	175.6	89.4	1.97	55,610
10	Office Uncategorized	State 6	1,160,832	234.2	73.6	3.18	54,630
11	Office Uncategorized	State 6	764,915	319.0	78.4	4.07	53,934
12	Health Care Uncategorized	State 6	925,705	316.9	137.1	2.31	48,781
13	Office Uncategorized	State 8	1,205,490	192.8	79.6	2.42	40,004
14	Office Uncategorized	State 6	948,747	212.6	76.1	2.79	37,957
15	Public Assembly Entertainment culture	State 4	1,008,416	248.7	126.3	1.97	36,194
16	Health Care Outpatient Uncategorized	State 6	745,000	370.0	217.4	1.70	33,319
17	Office Uncategorized	State 6	605,100	253.7	68.8	3.69	32,790
18	Service Vehicle service repair shop	State 6	759,046	234.7	97.3	2.41	30,547
19	Retail Enclosed mall	State 4	864,923	238.4	122.8	1.94	29,322
20	Health Care Inpatient	State 6	309,327	607.1	288.7	2.10	28,866
21	Education College or university	State 6	723,777	178.7	44.7	4.00	28,432
22	Office Uncategorized	State 6	466,063	278.0	73.4	3.79	27,950
23	Multifamily Uncategorized	State 6	1,511,780	150.9	89.9	1.68	27,024
24	Data Center	State 6	133,983	829.6	173.7	4.78	25,756
25	Commercial Other	State 11	535,947	244.5	83.2	2.94	25,328
26	Commercial Other	State 6	334,751	334.0	79.1	4.22	25,011
27	Office Uncategorized	State 6	275,533	390.5	101.3	3.86	23,359
28	Industrial	State 6	1,129,933	196.5	126.0	1.56	23,324
29	Education College or university	State 2	2,001,995	104.1	64.4	1.62	23,262
30	Lodging Hotel	State 4	515,556	266.1	112.9	2.36	23,145
31	Public Safety Penitentiary	State 8	436,250	266.6	90.3	2.95	22,545

Field	Value
Building type	Health Care Uncategorized
State	State 6
Floor area	3,032,242 ft2
Year built	1962
Energy Star rating	not rated
Excess energy	133,331 MWh/yr
Excess CO2 (assumed)	50,666 t/yr
Building id	61899

How to use it

1. Choose a list: retrofit candidates or meter checks.
2. Filter by building type and minimum ratio (1.5× to 5×).
3. The summary updates: count, excess GWh, CO_2 and median Energy Star.
4. Click a building to see its actual vs expected energy and details.

Who it is for: a facilities manager can start at the top of the list and see how far each building sits above its benchmark. The curve shows how much of the list's excess sits in its top 10%.

Impact: a ranked list of buildings to audit

2,973

retrofit candidates in Year 6 (13% of buildings), ranked in a CSV list

8,215

GWh a year of excess energy if every candidate matched its benchmark (an upper bound)

3,122

kilotonnes of CO_2 a year, at an assumed 0.38 t CO_2 per MWh

Also delivered: energy predictions for all 9,705 Year 7 buildings, 107 readings flagged for meter checks, and a median excess of about 1,305 MWh a year per candidate.

Limits, assumptions and next steps

Limits and key assumptions

- A screening tool, not an audit: a flag says which building to check, not what to fix.
- R^2 0.50: the data has no occupancy, opening hours or equipment information.
- CO_2 uses one assumed factor (0.38 t/MWh); savings are an upper bound.
- Years and states are anonymised, and State 6 skews the overall scores.

Next steps

1. Choose models per building type, or use the raw-target model for Year 7 (no State 6).
2. Add retrofit costs to rank by cost per tonne of CO_2 saved.
3. Re-run each year as new meter data arrives.
4. Add occupancy and operating-hours data where available.

Full report, code and app: github.com/chuaguanyu0718-dotcom/building-energy-prediction